# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

**Total Marks:50**

Annexure A

## APPLICATION- BASED QUESTIONS

***Code of Conduct:***

*I K.S. Dinesh (192324105) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

Dinesh

***Signature of the student:***

Annexure A

# APPLICATION- BASED QUESTIONS

1. **Scenario: Event Communication Security**
A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. **Scenario: Intercepted Suspicious Message**
A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
   a. Deduce the most likely original message.
   b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. **Scenario: Secure Messaging System**
A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. **Scenario: Suspicious Digital Asset Investigation**
During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. **Scenario: Legacy Encryption in Corporate Communication**
A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".
   a. Determine the most probable original message by reversing the encryption process.
   b. Critically evaluate why such a method is vulnerable in modern communication systems.
   c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

## RUBRICS FOR CALCULATION:

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25 | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design | 30 | Well-planned, systematic implementation with optimal solution | Correct implementation with minor issues | Basic implementation with errors or inefficiencies | Incorrect or incomplete implementation |
| Use of Tools | 20 | Effective and advanced use of tools/technologies with proper configuration | Appropriate use of tools with correct execution | Limited or inconsistent use of tools | Incorrect or minimal use of tools |
| Analysis of Results | 15 | Insightful analysis with accurate interpretation and validation of results | Good analysis with reasonable interpretation | Basic analysis with limited insights | Poor or incorrect analysis of results |
| Documentation | 10 | Clear, well-structured documentation with diagrams, code, and results | Organized documentation with necessary details | Basic documentation with missing details | Poor or incomplete documentation |

# Suspicious Digital Asset investigation

## Introduction:

In digital forensics, steganography is the technique of hiding secret information inside another file such as an image, audio or video.

a) How can investigators detect hidden data within the image:

Step: 1 Examine the image normally
* unusal visual Patterns.
* strange Colors or noise

Step: 2 check file Properties:
* File size
* image dimensions

Step: 3 Analyze the Least significant Bit (LSB)
A Common steganograph method hides information in the Least significant Bits (LSBs) of image Pixels.

step: 4 Perform statistical analysis:
* Pixel frequency
* NOise Patterns

Step: 5 use steganalysis tools:
* Detect Possible hidden data.

b) Two Techniques to strengthen security of hidden Communication:

1) Encrypt the data before hiding it:

The secret message should first be encrypted using a strong encryption algorithm as AES.

Process:

original Message → Encryption → ciphertext → Steganography → image

2) use advanced / adaptive steganography:

Instead of hidding data in Predictable location, adaptive steganography can select image areas with more natural noise.

* Edges
* Textured regions
* High Frequency areas

Conclusions:

Investigators can detect hidden information by examining file properties, metadata, pixel values, LSBs, statistical Patterns and image histograms, followed by forensic steganalysis tools.

Coded Message : GSRH RH HVXIVG

a) Deduce the original Message:

Step: 1

The message appears to use a simple substitution cipher.

Step: 2

Plain: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Z Y X W V U T S R Q P O N M L K J I H G F E D C B A

So.

A → Z
B → Y
C → X
G → T
H → S
R → I
S → H

Decode each word:

GSRH → G → T, S → H, R → I, H → S

∴ GSRH → THIS

RH → R → I, H → S RH → IS

Z → Z → A Z → A

HVXIVG :

H → S
V → E
X → C
I → R
V → E
G → T

HVXIVG → SECRET

∴ This is a secret

b) Justification:

Step 1 → check the letter Pattern:

4 letters → 2 letters → 1 letter → 6 letters

This matches:

This is a secret

Step 2 → check substitution Consistency:

R → I
H → S
V → E
G → T

step 3 → identify the known characteristic:

The mapping exactly match the Atbash cipher, where the alphabet is reversed.

Final Answer:

a) original message: This is a Secret

b) The cipher text uses the reversed alphabet.